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Paper Code : OECCS701B Multimedia Systems

UPID : 007563

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following :

[1 x 10 = 10]

- (I) What is tweened path animation ?
- (II) Which device is used to separate sound outputs picked up by separate microphones in a TV receiver or a DVD player in surround sound stereo receiver?
- (III) Name three hardware and software requirements for text element in multimedia.
- (IV) HTTP client accesses Web server by using the_____.
- (V) A frame buffer on a black and white system with one bit per pixel is commonly called _____.
- (VI) Name the file extensions of real audio.
- (VII) Moving Picture Experts Group (MPEG-2), was designed for high-quality DVD with a data rate of_____.
- (VIII) Multimedia is also used for _____ communications and presentations.
- (IX) In Joint Photographic Experts Group (JPEG), a gray scale picture is divided into blocks of_____ pixels.
- (X) RTP uses a temporary even-numbered_____.
- (XI) Audio compression can be used for_____.
- (XII) In Voice Over IP, Term SIP stands for_____.

Group-B (Short Answer Type Question)

Answer *any three* of the following :

[5 x 3 = 15]

2. What is meant by continuous tone, half-tone and bitone images? [5]
3. Write a short note: SGML [5]
4. How does the CMYK color model represent color information? Why is it called a subtractive model? [5]
5. Discuss DVD and CD-ROM. [5]
6. Discuss the components of multimedia . [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following :

[15 x 3 = 45]

7. (a) Write a short note on video conferencing. [5]
(b) Write a short note on video on-demand. [5]
8. (a) What are the basic steps in image processing? [5]
(b) What is a scanner? How does it function? [5]
(c) Discuss Image File Formats. [5]
9. (a) Discuss the Multimedia and Hypermedia Coding Expert Group (MHEG). [5]
(b) Write a short note on Open Document Architecture (ODA). [5]
(c) Discuss the various type of compression. [5]
10. (a) Discuss the fundamental characteristics of sound. [5]
(b) Define decibels. What is meant by saying that a sound has a loudness of 60 dB? [5]
(c) Explain the raster scanning process. With reference to the same context, explain "The refresh rate of a monitor is 60 Hz". [5]
11. (a) Define Sampling. What is meant by sampling rate? [5]
(b) Distinguish between analog and digital waves. [5]
(c) Write a note on Nyquist Theorem. [5]